॥ श्रीगणेशाय नमः ॥

Problem Solving

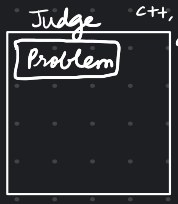
Brute force search

A brute force search (or exhaustive search or generate and test) is a problem-solving technique that consists of systematically checking all possible candidates for whether or not each candidate satisfies the problem's statement.
For a problem, a brute-force solution generally gives a worse time complexity.

Competitive programming

Online judge

An online judge consists of compilers in various language.



e.g. Problem:
Check if a # is prime / not.



The problem has an $O(n)$ solution.

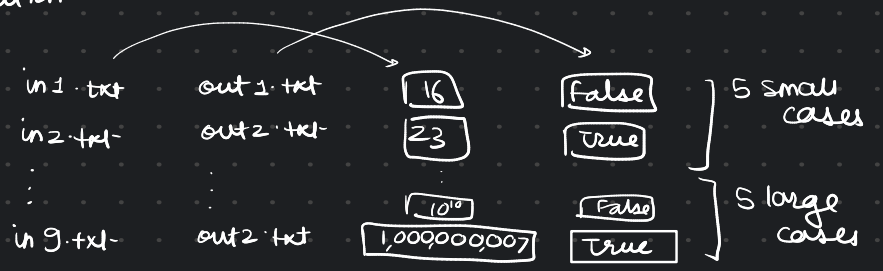
We could check $2, 3, \ldots, n-1$.
We could also check only upto $\frac{n}{2}$.
Say a k divides n s.t. $k \in (\frac{n}{2}, n)$
Then $\frac{n}{k}$ would also divide n.
That would be a better soln.

An even better solution could be to check till $\sqrt{n}$.
If $k > \sqrt{n}$ that divides n,
$\exists\ k_2 < \sqrt{n}$ that divides n.

$O(\sqrt{n})$ is a much better time complexity.

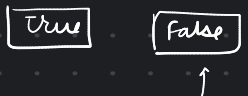
Say we're looking for an $O(n)$ solution.

Test files:
Every test file is an ordered pair of input and output files.



Say we want to test "temp.cpp"
we'll run

```
g++ temp.cpp -o temp
./temp < in.txt > result.txt
```



} verdict will be passed on comparison with out.txt (otherwise FAIL / PASS)

Say we actually return an $O(n)$ soln.
For $n = 1,000,000,007$

our soln:

```
for (int i=2 ; i<n ; i++) {
    if (n % i == 0)
        return false;
}
return true;
```

The judge would keep a time limit, say 1 sec to run the prog (usually diff. for diff. prog. languages).
time: C < C++ < Java < Py
(C is faster)

The CPUs we usually run these on (usually some web service), have a clock cycle
Say
~ 1 GHz
$\sim 10^9$ Hz
$\sim 10^9$ operations/sec

Say a 4 line code is converted to 10 lines of assembly.
So for input 10^9, we'll need 10^9 operations, i.e. 10 secs.

If an $O(n)$ solution is expected, a time limit would be set to that.
Else, a TLE verdict would be passed.
↳ time limit exceeded.

A program would generally run $10^6 - 10^8$ opera^{n}s.

If $n \sim 10^6$, an order n soln would work.

Ly, an order n^2 soln would probably not work for an input size of $\sim 10^6$.
It would work, say for some input 10^3.

Other verdicts:
segmentation fault: The compiler fails at runtime, trying to do stuff not allowed [say accessing memory that isn't allowed]
compiler errors;

There are similar memory constraints placed. Resources are constrained.